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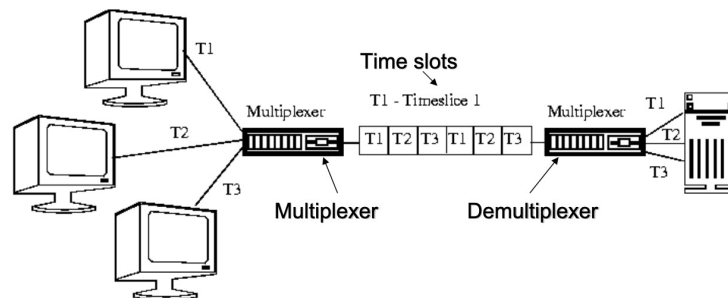
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Time Division Multiplexing

- *Time division multiplexing (TDM)* combines several low, fixed and predefined bit rate sources into a single high speed bit stream for transmission over a single digital communication channel:

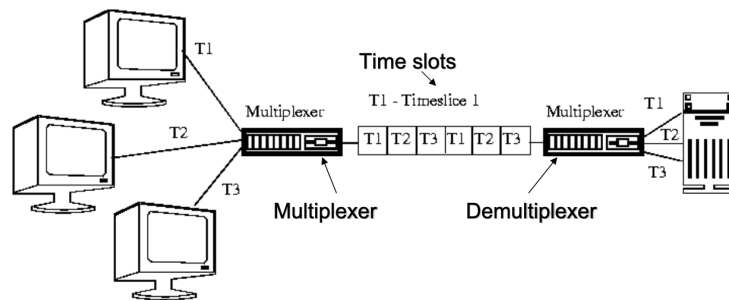


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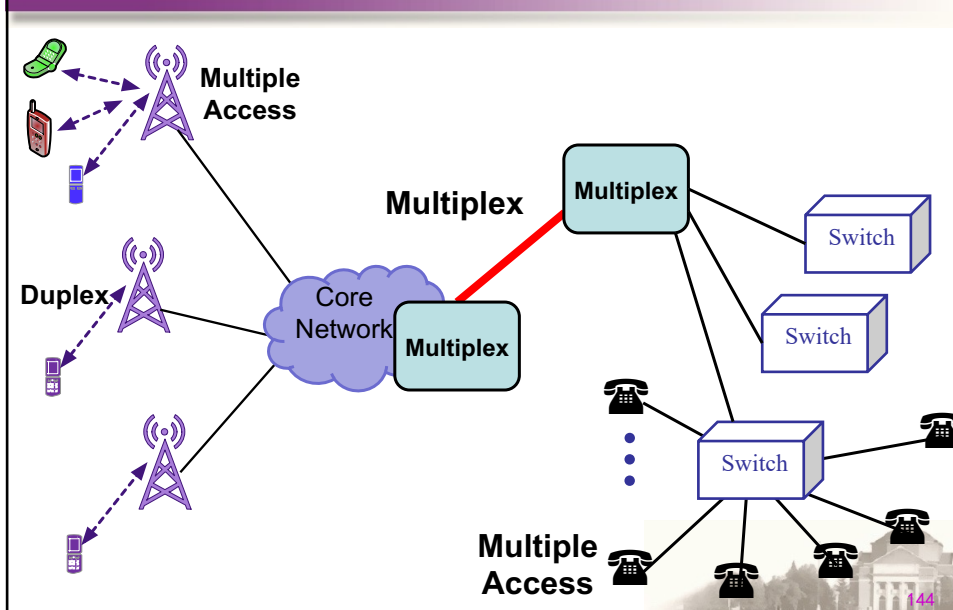
- The TDM time slots have to be chosen properly. If the time slots are too small (for example, 1 bit) then the multiplexer and demultiplexer must switch rapidly. If the time slots are too large (for example, 64 Kb) then the data must be *buffered* and *delay* would be produced. TDM is used for baseband (*not* bandpass) data transmission.



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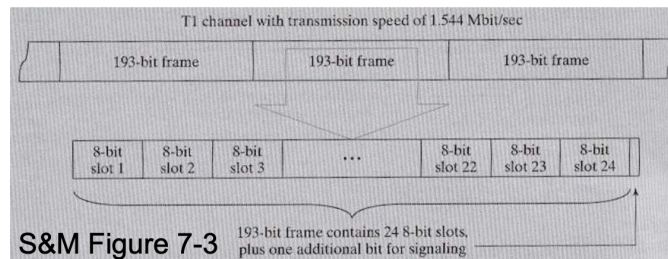
Oversea Calls



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- The T1 TDM system for telephone networks uses a 193 bit *frame* where each frame has 24 8-bit *slots* and 1-bit in each frame for signaling. The T1 bit rate $r_b = 1.544$ Mb/sec.



The duration of each frame $T_f = 193 \text{ b} / 1.544 \text{ Mb/sec} = 125 \mu\text{sec}$ or 8 k samples/sec. Here all the data sources have the same data rate.

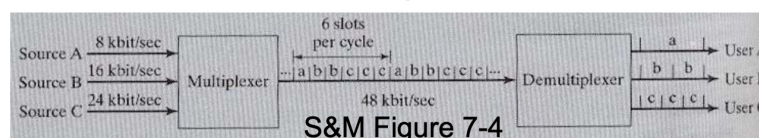
S&M: Communication Systems analysis and design,
Harold P.E. Stern, Samy Mahmoud, Prentice-Hall, Inc.

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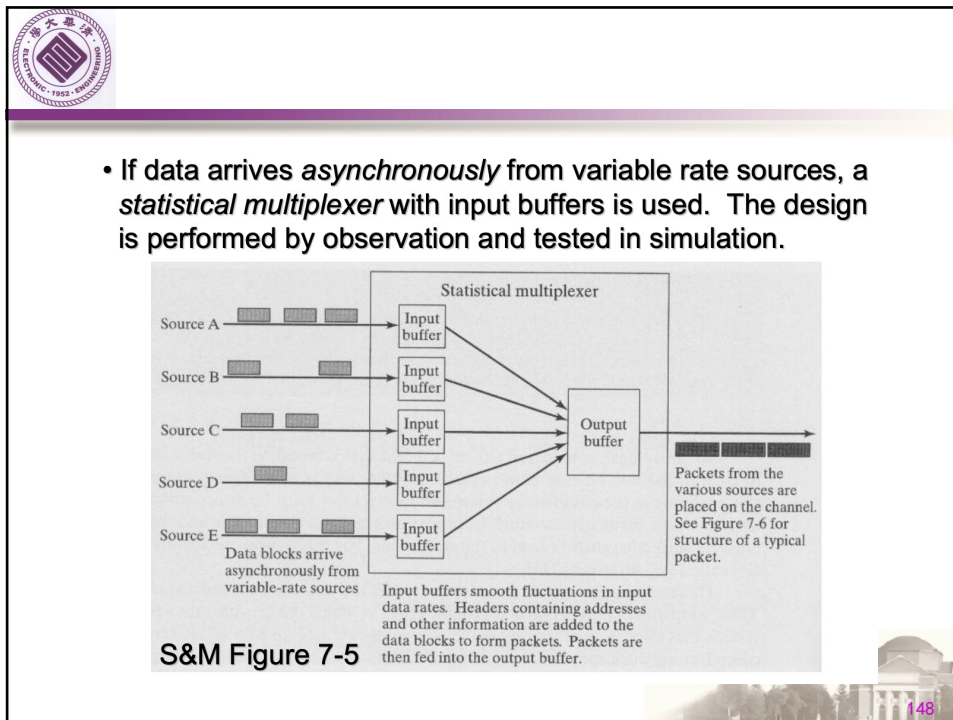
- If the data sources have different rates a *multiplexer scheme* must reconcile the disparate rates.



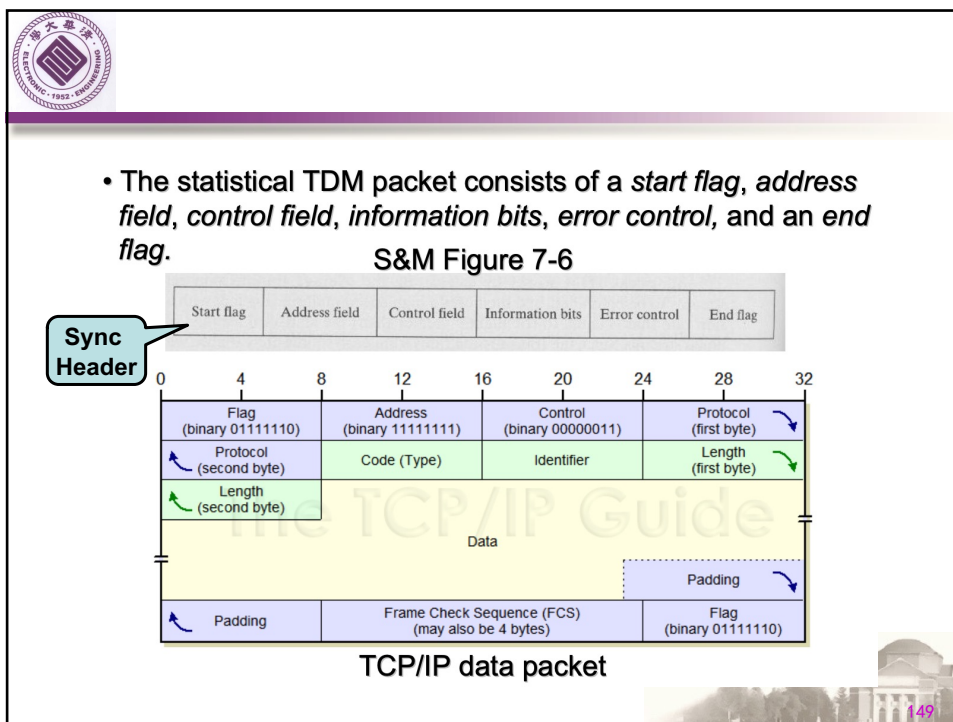
The data rates are in the lowest possible ratio of 8:16:24 or 1:2:3 for a total of $1 + 2 + 3 = 6$ slots. The slots are apportioned to the channels as: *a b b c c c* and the channel data rate is 48 kb/sec.

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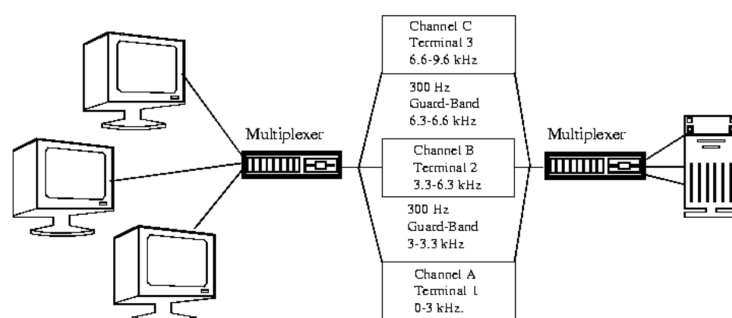
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- **Frequency division multiplexing (FDM)** divides the total bandwidth available to the system into *non-overlapping* frequency sub-bands for transmission over a single digital communication channel.

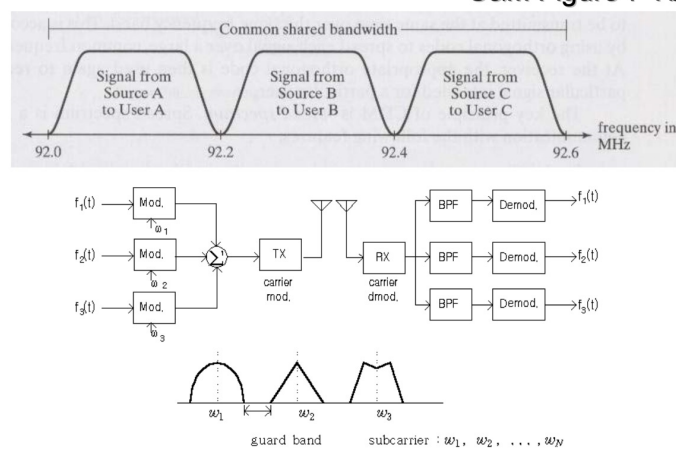


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- FDM employs a common shared bandwidth but here with no apparent guard bands.

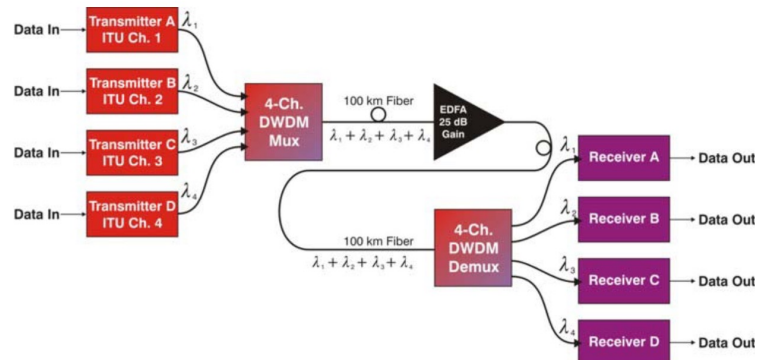
S&M Figure 7-7b



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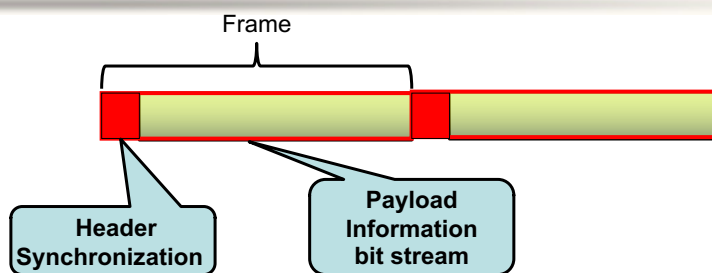
- Optical FDM is usually called *wavelength division multiplexing* (WDM) and utilizes separate wavelengths (λ) of light.



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Demultiplexing



a special bit pattern in header
sentinel is 01111110 (0x7E)

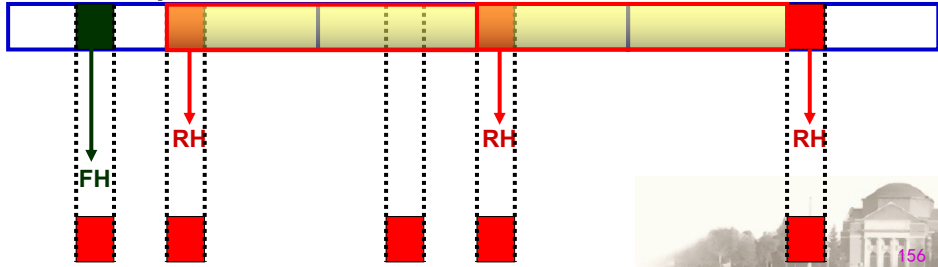
- ⊕ Fake-header: Payload may also show the sync pattern;
- ⊕ Missing the header: bit error in transmitting.

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Off-line scheme

- ⊕ Save all bit stream
- ⊕ Search the sync patter
 - ❑ the frame length is fixed
 - ❑ Confirm/Catching Real-Header (RH), when get N consecutive patters
 - ❑ Split the frames

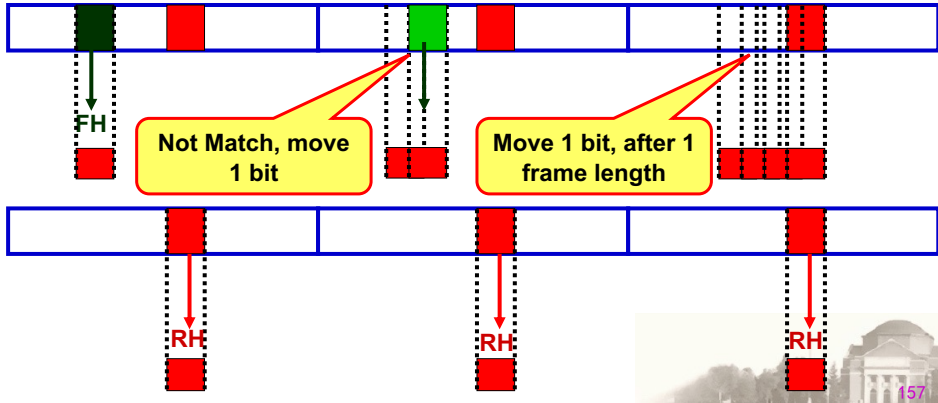


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On-line scheme

- ⊕ No storage
- ⊕ Move 1 bit, if not match the patter in next position



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Sync capture time (Optional)

- Header pattern length is n , Probability of FH is

$$P = 2^{-n}$$

- If FH is 1 bit ahead of RH, we need ΔT time to confirm it is FH

□ T is one frame time, T_h is one bit time,

□ There are i consecutive Fake-Headers in bit stream, so we could confirm it in $(i+1)$ frame position

$$\begin{aligned}\Delta T &= \sum_{i=1}^{\infty} P^i (1-P) i T \\ &= (1-P) T \sum_{i=1}^{\infty} i P^i \\ &= P(1-P)^{-1} T\end{aligned}$$

If FH is k bit ahead of RH

$$T_k = (k-1)\Delta T + (k-1)T_h$$

Each position may be FH (P)

Time cost to shift 1 bit position

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Sync capture time (Optional)

- Bit stream is split randomly, then the time to find the first Real-Header is expectation of T_k

□ Frame length is L ,

$$\begin{aligned}\bar{T} &= E[T_k] \\ &= \sum_{k=1}^L \frac{(k-1)\Delta T + (k-1)T_h}{L} \\ &= \frac{L-1}{2} (\Delta T + T_h) \\ &= \frac{L-1}{2} \left(\frac{P}{1-P} T + T_h \right)\end{aligned}$$

$T = L \cdot T_h$

$$\begin{aligned}\bar{T} &= \frac{L-1}{2} \left(\frac{P}{1-P} T + T_h \right) \\ &= \frac{L-1}{2} \left(1 + \frac{LP}{1-P} \right) T_h\end{aligned}$$

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Sync capture time w/ bit error (Optional)

- Miss the RH, the probability is, (BER= P_e)

$$P_m = 1 - (1 - P_e)^n \approx nP_e$$

Miss a 'RH'



- Need to search RH from next bit position, it costs $T_L + T_h$ time to find next RH.

- Average time to catch RH is,

There are i consecutive bit-error RH missed in capture process.

$$\begin{aligned} \bar{T}_e &= \sum_{i=0}^{\infty} P_m^i (1 - P_m) [\bar{T} + i(T_L + T_h)] \\ &= \bar{T} + (1 - P_m)(T_L + T_h) \sum_{i=1}^{\infty} iP_m^i \\ &= \bar{T} + \frac{P_m}{1 - P_m} (T_L + T_h) \end{aligned}$$

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Sync capture time with bit error (Optional)

$$\begin{aligned} \bar{T}_e &= \sum_{i=0}^{\infty} P_m^i (1 - P_m) [\bar{T} + i(T_L + T_h)] \\ &= \bar{T} + (1 - P_m)(T_L + T_h) \sum_{i=1}^{\infty} iP_m^i \\ &= \bar{T} + \frac{P_m}{1 - P_m} (T_L + T_h) \end{aligned}$$

$$2\bar{T} = (L - 1)(\Delta T + T_h) \approx T_L + T_h$$

$$\bar{T}_e \approx \frac{1 + P_m}{1 - P_m} \bar{T} \approx (1 + 2P_m)(1 + LP) \frac{L - 1}{2} T_h \approx (1 + 2P_m + LP) \frac{L - 1}{2} T_h$$

P and P_m are very small

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Optimal length of header (Optional)

$$\bar{T}_e \approx (1 + 2P_m + LP) \frac{L-1}{2} T_h$$

$$P_m = nP_e$$

Short header

Long header

$$P = 2^{-n}$$

$$\min : t(n) = 2nP_e + L2^{-n}$$

$$\frac{d}{dn} t(n) = 2P_e - L2^{-n} \ln 2 = 0$$

$$\text{Opt: } n = \log_2 \left(\frac{L \ln 2}{2P_e} \right)$$

T1-TDM: 24*64kbps (7/8), 1.544Mbps
n=25 bit, L=193, Th=128uS/193
Pe=3e-6

Opt: n=24.4,
Te~0.5(frame time)



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Thanks !



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